WI4465 Advanced Topics in Probability.
Fall 2026/27

PROJECT 3: Does clustering change the 3-core phase transition?
3-cores in random intersection graphs
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The k-core of a graph G is the maximal induced subgraph in which every vertex has degree at least k. It can be found algorithmically by repeatedly deleting all vertices whose current degree is smaller than k, until no such vertex remains. The final graph does not depend on the order in which vertices are deleted.

The k-core undergoes an interesting phase transition in classical random graphs. For instance, for $k \geq 3$ the k-core of an Erdős–Rényi graph suddenly becomes linear in size at a critical average degree. The k-core of the configuration model with a general degree distribution is also well understood, see [1].

In this project we ask what happens if the graph has a lot of *clustering*. We use a random intersection graph (RIG): individuals belong to small groups, and all members of the same group are connected to each other. Thus each group produces a clique, and different cliques overlap through individuals who belong to several groups. This is a special case of the random intersection graph with communities introduced in [2,3].

The simplest interesting case for the 3-core is when every group has exactly 3 members, so every community is a triangle. A single triangle is *not* itself a 3-core, since each of its vertices has only two neighbors inside the triangle. Thus, if a non-trivial 3-core appears, it has to arise from several overlapping triangles supporting each other. This makes the triangle model a particularly clean model for studying the effect of clustering.

Some papers to start with. You do not need to read all proofs! Try first to understand the models, the main results and the ideas behind the proofs.

[1] Janson, S. and Łuczak, M.J. (2007) A simple solution to the k-core problem. Random Structures & Algorithms 30, 50–62. Online here

[2] van der Hofstad, R., Komjáthy, J. and Vadon, V. (2021) Random intersection graphs with communities. Advances in Applied Probability 53, 1061–1089. Online here

[3] van der Hofstad, R., Komjáthy, J. and Vadon, V. (2022) Phase transition in random intersection graphs with communities. Random Structures & Algorithms 60, 406–461. Online here

[4] Gleeson, J.P. and Melnik, S. (2009) Analytical results for bond percolation and k-core sizes on clustered networks. Physical Review E 80, 046121. Online here

1. Start with the triangle random intersection graph.

There are n individuals. Give individual i a number D_i of community memberships, where for the simplest version you can take

$$D_i \sim \mathrm{Poi}(\lambda) \tag{1}$$

independently. Give individual i exactly D_i membership half-edges. On the other side create community vertices, each with exactly 3 half-edges, and match the two sets of half-edges uniformly (you may need to make a tiny adjustment so that the total number is divisible by 3). Thus each community has three membership slots. After the matching, connect every pair of individuals that belong to the same community. Usually the three slots contain three different individuals and hence form a triangle; occasionally the same individual may occupy more than one slot, which may create loops or multiple edges. You may simply keep these. Equivalently, this is a bipartite configuration model in which all community-side degrees equal 3, followed by projection onto the individual side; see [2].

Write a simulation of this graph and an efficient algorithm for finding its 3-core. Vary λ and plot

$$\frac{|\mathrm{Core}_3(G)|}{n} \tag{2}$$

as a function of λ. Average over several graphs and increase n. Do you see a critical value λ_c? Does the 3-core seem to appear continuously from size 0, or does it make a jump to a positive fraction of the graph?

A useful comparison is the configuration model with (approximately) the same degree distribution. In the triangle RIG, an individual with D memberships has approximately degree $2D$ (ignoring the rare event that the same neighbor is met in two groups). Thus compare with a configuration model having degree distribution

$$D_{\mathrm{CM}} = 2D, \qquad D \sim \mathrm{Poi}(\lambda). \tag{3}$$

Read enough of [1] to understand how the 3-core transition of this configuration model can be predicted. Compare the two critical values and the sizes of the cores. What is genuinely caused by clustering, rather than by the degree distribution?

2. A non-rigorous cavity/tree recursion for the triangle model.

The projected graph has lots of triangles, so it is certainly not locally tree-like as an ordinary graph. However, its underlying *bipartite* graph of individuals and groups is locally tree-like. This suggests a cavity/message-passing approximation: remove one individual–group incidence and pretend that the messages arriving through the other groups are independent.

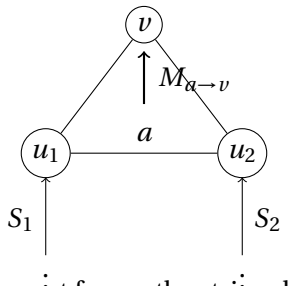
Consider one triangle a containing the vertices v, u_1, u_2, and think of v as the parent/root. Suppose for the moment that v itself stays in the 3-core. For $i = 1, 2$, let

$$S_i = \text{number of surviving neighbors that } u_i \text{ gets from triangles other than } a. \tag{4}$$

Thus S_i measures the *external support* of u_i. The triangle a may then contribute $0, 1$, or 2 surviving neighbors to v. Let

$$M_{a\to v} \in \{0, 1, 2\} \tag{5}$$

denote this number.



Think of v as the parent. The quantities S_1, S_2 summarize everything happening below u_1, u_2; the triangle a then sends the message $M_{a\to v}$ upward to v.

Can you determine $M_{a\to v}$ from (S_1, S_2)? There are only three cases:

- If $S_1 \geq 1$ and $S_2 \geq 1$, then both u_1, u_2 survive, since each has the two neighbors inside the triangle plus at least one external surviving neighbor. Hence $M_{a\to v} = 2$.
- If, say, $S_2 = 0$, then u_2 cannot survive. In that case u_1 has only v left inside the triangle, so it survives precisely when $S_1 \geq 2$. This gives $M_{a\to v} = 1$. (And symmetrically with 1 and 2 interchanged.)
- In the remaining cases $M_{a\to v} = 0$.

Let

$$q_j = \mathbb{P}(M = j), \qquad j = 0, 1, 2, \tag{6}$$

and let S have the same distribution as the external support S_i. The local triangle calculation gives

$$q_2 = \mathbb{P}(S \geq 1)^2, \qquad q_1 = 2\mathbb{P}(S = 0)\mathbb{P}(S \geq 2). \tag{7}$$

Now comes the recursive part. The external support S of a vertex is itself obtained by adding the messages coming from all its other triangles. If $D \sim \mathrm{Poi}(\lambda)$, then after following one membership edge the number of remaining memberships is again $\mathrm{Poi}(\lambda)$. Hence the tree approximation predicts

$$S \overset{d}{=} \sum_{i=1}^{N} M_i, \qquad N \sim \mathrm{Poi}(\lambda), \tag{8}$$

where the M_i are iid with $\mathbb{P}(M_i = j) = q_j$.

Can you close these equations? As a hint, if $p_j = \mathbb{P}(S = j)$, then

$$q_2 = (1 - p_0)^2, \qquad q_1 = 2p_0(1 - p_0 - p_1), \tag{9}$$

and the compound Poisson distribution in (8) gives

$$p_0 = \exp\{-\lambda(q_1 + q_2)\}, \qquad p_1 = \lambda q_1 p_0. \tag{10}$$

Compute also p_2 and obtain a closed finite-dimensional fixed-point problem.

Solve/iterate these equations numerically as λ varies. A vertex belongs to the 3-core if the total support from all its triangles is at least 3, so the predicted core density is

$$\rho_3(\lambda) = \mathbb{P}(S \geq 3) = 1 - p_0 - p_1 - p_2. \tag{11}$$

Compare (11) with your simulations. Can you locate the critical point from the fixed-point equations? Does the non-zero solution appear continuously or discontinuously?

This calculation is intentionally *not* presented as a proof. Try to explain carefully what independence/tree-like assumptions are being made, and why they are plausible on the bipartite graph even though the projected graph contains many triangles. How accurate are the predictions for finite n?

3. Can you understand the recursion probabilistically?

The usual configuration-model k-core analysis in [1] follows the pruning process and can be encoded using a continuous-time death process. Here the pruning is less local on the bipartite graph: deleting one individual changes the projected degrees of all other individuals in the same groups, i.e., vertices at distance two in the bipartite graph. So you should *not* expect the proof in [1] to carry over word for word.

Instead, try to interpret the cavity recursion as a "warning propagation" algorithm on the locally tree-like factor graph. Starting from all messages active, repeatedly update the messages from individuals to groups and from groups to individuals according to whether enough neighbors can still survive. Does this numerical message-passing algorithm converge to the same answer as direct 3-core pruning on large graphs?

If you like theory, you can think about what would be needed to turn this heuristic into a theorem: [local weak convergence alone describes every fixed neighborhood, whereas the 3-core pruning may take many rounds. A clear discussion of the difficulty is already useful.]

4. What changes when communities can be larger?

Now allow group sizes bigger than 3, still replacing every group by a complete graph. This introduces a new phenomenon which is special to intersection graphs. A clique of size $s \geq 4$ is already a 3-core by itself, since every vertex has at least 3 neighbors inside the clique. Therefore, if there are linearly many such communities, the total 3-core may be linear for a completely local/trivial reason, even before any global phase transition occurs.

Choose, for example, a mixture in which communities are triangles with probability $1-r$ and K_4's with probability r, and vary λ and/or r. Some possible questions are:

- How large is the whole 3-core, and how much of it is already explained by the union of the K_4 communities?
- Does the *largest connected component of the* 3*-core* undergo a phase transition, even though the total 3-core is already linear?
- Do the automatically surviving K_4's recruit additional vertices/triangles into the 3-core?
- What happens if the community-size distribution has several sizes, or a heavy tail?

You can also try to generalise the cavity idea. If a community has size s, the message from this community to a distinguished vertex can now take values in $\{0,1,\ldots,s-1\}$. Given the external supports of the other $s-1$ members, determine how many of them can survive together inside the clique. For $s=4$ this can still be done by hand; for

larger s you can implement the local pruning rule algorithmically and use it inside the fixed-point iteration.

A good project does not need to answer all of these questions. One route is a careful simulation study plus the cavity equations for the pure triangle model. A more theoretical group can focus on deriving and understanding the fixed-point system and comparing it with [1]. An exploratory group can investigate mixtures of clique sizes and formulate a conjecture about the giant connected 3-core.

Try to combine simulations with at least one probabilistic calculation or heuristic argument, and be explicit about finite-size effects and about which conclusions are rigorous and which are conjectural.

Prepare a presentation on the studied results and your findings.

What to avoid.

- Mishandle repeated memberships, loops, parallel edges, or the comparison degree distribution.
- Implement the cavity equations without understanding which message or support distribution is being updated.
- Find only the zero fixed point or miss a discontinuous branch because of initialization.
- Extend immediately to K_4 communities and then mistake the automatic local 3-cores for a global transition.